

Platform and Test Engineer (Developer 2)

CAM/AUTO | Mumbai, on site | Full time | Job description, 29 September 2026

Company	CAM/AUTO, the in-house CAM software of a precision CNC machine shop
Location	Mumbai, on site. The role owns the shop-floor measurement loop and the on-site server, so it cannot be remote
Type	Full time. Six-month go-live plan with a completion bonus at the week-21 gate; ongoing role in phase two
Reports to	Founder (product owner). Works daily with the geometry lead and the process owner (senior CNC programmer)
Compensation	INR 18 to 28 lakh a year (Mumbai 2026 band, to be confirmed), plus a completion bonus tied to the week-21 first-article gate

The role in one paragraph

You will turn a working single-file CAM prototype into an engineered product without stopping it. Today the app is 12,600 lines of JavaScript in one file with no continuous integration, a Node test harness of 14 suites, and a six-part release gate that runs by hand. You will split it into typed modules, put every commit behind CI with a real part gate, build the measurement loop that brings CMM results back from the shop floor against the program that produced them, and run the on-site server that holds it all. The learning module cannot exist without the data you collect; the geometry lead cannot prove exactness without the tests you run. Your gate is week 3: CI green and the accuracy baseline measured on six real parts.

What you will do

Before week 1, with the founder (if you can start early)

- Take over the mechanical module split that the founder and Claude Code have begun; decide the module boundaries with the geometry lead; get the build producing the same single HTML file the shop runs today.

Weeks 1 to 3: foundation

- CI on every commit on a self-hosted GitHub Actions runner on the on-site server: syntax, the 14 suites, the smoke run, the Playwright browser run and the part gate. Nothing merges red.
- Start the TypeScript migration one module at a time, strict mode on new code; esbuild bundle into the single file.
- Build the corpus: golden outputs (recognised features, paths, posted NC) per trial part, stored on the on-site runner only; customer files never enter git.
- Program identity: a SHA-256 hash of inputs, tools, machine and post, printed on the setup card and carried by every NC file.
- Measurement form and CMM CSV import; record today's accuracy on the six trial parts as the KPI baseline.

Weeks 3 to 15: second check and posts

- Test infrastructure for the geometry lead's exact recognition and offsets: property-based tests, comparisons against the STEP file, tolerance reports per part.
- Integrate the independent NC re-simulation into CI so every posted program is checked twice before it is released.
- Run the post-processor validation: dry runs of Haas/Fanuc (and Siemens in the full scope) programs on the shop machines with the process owner, logged against program hash.

Weeks 12 to 26: shop loop, learning data and go-live

- On-site service (Node/Fastify or Python/FastAPI) with PostgreSQL for programs, measurements and history; Docker Compose on the on-site server; encrypted backups on the NAS and off site.
- Tablet entry page for the shop floor (full scope): setup card, tool list, measurement entry, tool-wear events.
- The learning data pipeline: measurement records joined to program hash, feature, tool, machine, fixture and operator, exported for the learning module; the offline corpus replay that gates every learned value.
- Operator training material with the process owner; hardening; the three setter-run jobs in month 6.

Technical skills and technologies

Core: must have, tested in the paid task

Area	Technologies	Level	Used for
TypeScript / JavaScript	ES2020+, TypeScript strict mode, ES modules, esbuild, Node 20, npm	Expert	Module split, migration, the bundle the shop runs
Test automation	Node test runner or Vitest, golden-file and snapshot tests, property-based testing (fast-check), Playwright end-to-end in Chromium, coverage reporting	Expert	The regression contract: 14 suites today, 30-part gate at go-live
CI/CD	GitHub Actions, self-hosted runners on Linux, branch protection, required checks, caching, artifact retention	Expert	Every commit gated; part files stay on site
Python	Python 3.11, pandas, the existing Python builder script, pytest	Strong	Builder, CSV import, learning data pipeline, corpus replay

Area	Technologies	Level	Used for
Linux and servers	Ubuntu/Debian administration, Docker and Docker Compose, systemd, nginx, UPS and backup routines, ZFS or mdadm mirrors	Strong	The on-site application and CI server
Databases	PostgreSQL (schema design, migrations, backups), SQLite; JSON schema validation	Strong	Programs, measurements, history
Backend	Fastify (Node) or FastAPI (Python), REST, file uploads, auth for two roles (admin, operator)	Strong	On-site service and tablet page
Git	Git, GitHub pull-request review, release tagging, reproducible builds	Strong	Traceability of every released program to a commit

Domain: strongly preferred, learnable in the first month

Area	Technologies	Level
CNC basics	Reads a G-code program (Fanuc/Haas ISO, Siemens); understands work offsets, tool tables, setup sheets; can stand a dry run with the process owner	Working
Metrology data	CMM report formats (CSV, PDF export), gauge readings, drawing tolerances; joins them to features and programs	Working
Geometry testing	Comparing paths and features against STEP data with tolerances; not writing geometry, but proving it	Working
Front end	Plain DOM and TypeScript for the tablet page; three.js awareness for the viewer build; no framework	Working
AI-assisted development	Claude Code: writes tasks with acceptance tests, reviews generated diffs, owns the token budget review	Working

Not required

Cloud platforms (everything runs on site); Kubernetes; machine learning frameworks (the learning models belong to developer 3; you own the data they run on); CAM or CAD kernel programming (the geometry lead owns it).

Experience profile

- 4 to 7 years as a platform, DevOps-minded or test-automation engineer on a product that shipped to real users, with at least one migration of a legacy codebase you did not write.
- Has built a CI pipeline and a regression harness from nothing and kept them green under a team.
- Comfortable on a factory floor: will spend one day a week with the process owner at the machine and the CMM.
- Likely backgrounds: product start-ups with a strong test culture, industrial software or IoT teams, SDET or platform roles at engineering-software companies.
- Degree in computer science or engineering, or equivalent shipped work.

How we hire

1. 30-minute call with the founder: the plan, the week-3 gate, your questions.
2. **Paid one-day task** on real repository material: take a 1,500-line slice of the current single file, split it into two ES modules with unit tests, and put it behind a GitHub Actions workflow that runs the tests and a Playwright smoke test on a self-hosted runner you configure in a container. We pay for the day whether or not we continue.
3. Two-hour review of your task with the founder and the geometry lead (or an external reviewer until the lead joins), then an hour at the shop with the process owner.
4. Reference calls. Offer within a week of step 3.

What you get

- A real product with 79 releases of shop findings and a shop floor that will cut the programs your pipeline releases.
- Ownership of the whole delivery path: build, tests, CI, server, data. No committee.
- Claude Code on your desk, an on-site workstation (16 cores, 64 GB), the on-site server to build as you see fit.
- Completion bonus at the week-21 gate and a phase-two role running the platform as the learning module grows.

Keywords: TypeScript, JavaScript, Node.js, esbuild, Vitest, Playwright, property-based testing, golden tests, GitHub Actions, self-hosted runner, CI/CD, Docker Compose, Linux, PostgreSQL, SQLite, Fastify, FastAPI, Python, pandas, legacy migration, monolith split, test automation, SDET, platform engineer, CNC, G-code, CMM, manufacturing software, on-premises.